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Editor-in-Chief
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mdpi

Dear Editor-in-Chief,

On behalf of my co-authors, I am pleased to submit our manuscript entitled “**A Microservice-Oriented Cyber Deception Platform with Containerized Honeypots and Real-Time Telemetry**” for consideration for publication in *Journal of Cybersecurity and Privacy*.

This paper addresses a key challenge in future and large-scale computing environments: the deployment and management of cyber deception infrastructures that are scalable, observable, and responsive under dynamic adversarial conditions. While cyber deception has proven effective for threat intelligence and attack diversion, existing platforms are often monolithic, statically configured, and poorly suited to modern cloud-native and distributed systems. These limitations restrict their applicability in next-generation computing environments characterised by elasticity, automation, and high concurrency.

To address this gap, we propose a microservice-oriented cyber deception platform that leverages containerisation, API-driven orchestration, and real-time telemetry. The platform decomposes deception functionality into independent services, enabling rapid deployment, elastic scaling, and fine-grained resource control. Containerised high-interaction honeypots are dynamically orchestrated through a lightweight control plane, while a centralised telemetry pipeline provides continuous, high-throughput behavioural data collection and monitoring. In addition, we introduce formal analytical models to characterise deployment latency, resource utilisation, telemetry throughput, and operational cost, providing principled insight into system scalability and performance.

The main contributions of this work include:

1. A microservice-based architecture for scalable and cloud-native cyber deception platforms.
2. An API-driven orchestration mechanism for rapid lifecycle management of containerised honeypots.
3. An integrated real-time telemetry and observability pipeline supporting high-throughput behavioural data collection.
4. Formal analytical models for deployment latency, scalability, and operational cost.
5. An empirical evaluation in a controlled cyber-range environment demonstrating low deployment latency, predictable resource scaling, and reliable telemetry ingestion under concurrent attacker workloads.

We believe this work aligns strongly with the aims and scope of journal of Cybersecurity and privacy. The proposed platform provides a practical and extensible foundation for future autonomous and AI-driven cyber defence systems deployed in large-scale and heterogeneous environments.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the manuscript and agree with its submission to *Journal of Cybersecurity and Privacy*.

Thank you for your time and consideration. We would be grateful for the opportunity to have our work reviewed and look forward to your response.

Yours sincerely,

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On behalf of all co-authors